

Get Fit with MA?

Estimating the Impact of Medicare Advantage on Physical Activity

Jong Min Jung*

Jing Li[†]

Abstract

Medicare Advantage (MA) plans increasingly offer supplemental fitness benefits, yet evidence on whether they change preventive health behavior among older adults is limited. Using nationally representative panel data from the Health and Retirement Study, this paper estimates the effect of MA enrollment on physical activity using a difference-in-discontinuities design that exploits the administrative Medicare eligibility threshold at age 65. MA enrollment increases the probability of daily vigorous exercise by about 3.9 percentage points relative to those not enrolled in MA. Event study estimates show no differential pre-trends. Effects are larger among women, non-White beneficiaries, and those without a college degree. These findings suggest that insurance benefit design can promote preventive health behavior at least in the short run.

JEL Classification: I12, I13, I18, J14

Keywords: Medicare Advantage, physical activity, preventive health behavior, older adults, difference-in-discontinuities

*Department of Economics, University of Washington

[†]Department of Pharmacy, University of Washington

1 Introduction

Physical activity plays a central role in maintaining health and independence in later life. Regular exercise reduces the risk of cardiovascular disease, diabetes, functional limitation, cognitive decline, and falls among older adults (Warburton et al. (2006); Lee et al. (2012)). Accordingly, the 2018 *Physical Activity Guidelines for Americans* recommend that adults over 65 engage in 150 - 300 minutes per week of moderate-intensity aerobic activity or 75 - 150 minutes per week of vigorous-intensity aerobic activity, along with multicomponent exercises that include muscle-strengthening training (U.S. Department of Health and Human Services (2018)). Despite the well-documented health benefits, physical inactivity remains widespread. In 2022, only 13.9 percent of U.S. older adults met federal physical activity guidelines (Centers for Disease Control and Prevention (2024); Elgaddal and Kramarow (2024)).

As the financial and logistical costs of exercise are potentially modifiable, health insurance may provide one channel for encouraging physical activity among older adults. Medicare Advantage (MA), also known as Medicare Part C, is a privately administered alternative to Traditional Medicare (TM) that provides Medicare Part A and Part B coverage through private insurers and includes supplemental benefits such as dental, vision, and fitness services that are not covered by TM. This makes MA a particularly relevant policy setting to investigate whether insurance design affects physical activity. MA enrollment now accounts for more than half of all Medicare beneficiaries, and by 2025, approximately 95 percent of individual MA plans offered a fitness benefit (Ochieng et al. (2025)). Fitness benefits are among the most common supplemental benefits and are provided through programs such as SilverSneakers, Renew Active, and Silver&Fit. These programs offer older adults free or subsidized access to gyms, group exercise classes, and community-based fitness opportunities. By lowering the financial and logistical barriers to exercise, MA plans may affect not only medical care use but also preventive health behaviors that shape later-life health.

Despite the rapid expansion of MA enrollment, causal evidence on its behavioral impact remains limited. Existing studies of MA-covered fitness program document improvements in participants' physical and mental health and reductions in health care spending, but the participants are likely to differ systematically from non-participants in motivation, baseline health, and exercise preferences (Belza et al. (2006); Hamar et al. (2013); Brady et al. (2020); Nguyen et al. (2008)). Thus, it remains unclear whether enrollment in MA changes exercise behavior around Medicare eligibility and, if so, whether the widespread availability of sup-

plemental fitness benefits represents one potential channel.

To address the selection problem, we use nationally representative panel data from the Health and Retirement Study (HRS) from 2012 to 2022 and exploit the Medicare eligibility threshold at age 65 as an exogenous source of variation. The main empirical strategy is a Difference-in-Discontinuities (DiDC) design that compares the age-65 change in daily vigorous exercise among MA enrollees with the corresponding change for non-MA individuals. The DiDC design differences out age-65 change common to both groups, such as retirement, Social Security claiming, and general aging-related trends, while isolating the MA-specific discontinuity in exercise behavior. Importantly, benefit availability does not imply actual participation and as we do not observe actual fitness-program uptake or utilization, our empirical analysis estimates the reduced-form behavioral effect of MA enrollment. We find that MA enrollment increases the probability of daily vigorous exercise by approximately 3.9 percentage points relative to the corresponding changes in non-MA group. The effects are larger among women, non-White beneficiaries, and individuals without a college degree, consistent with the idea that supplemental fitness benefits may matter most for groups facing greater barriers to exercise. Event study estimates show no evidence of differential trends prior to age 65, while the effect is concentrated near the eligibility threshold and attenuates over the following two years. This pattern suggests that access to fitness benefits may generate an initial behavioral response but may not lead to sustained increases in exercise on its own.

This paper makes two main contributions. First, to our knowledge, it provides the first quasi-experimental evidence on the reduced-form effect of Medicare Advantage enrollment on physical activity around the Medicare eligibility threshold. In addition, this paper contributes to the literature on insurance benefit design and preventive health behavior. Prior evidence on financial incentives for exercise behavior is largely concentrated in the experimental or workplace settings among younger adults (Charness and Gneezy (2009); Acland and Levy (2015); Royer et al. (2015)). We extend this literature to older adults and to a large public insurance setting, where incentives are embedded in plan benefits rather than offered as short-term experimental payments. Our findings suggest that insurance benefit design can induce preventive health behavior among older adults.

2 Background

2.1 Medicare and Medicare Advantage

Medicare is the federal health insurance program that primarily covers adults age 65 and older, as well as some younger individuals with disabilities or end-stage renal disease. Most beneficiaries become eligible at age 65, creating a major transition in insurance coverage and healthcare financing. Traditional Medicare, also called fee-for-service Medicare, consists principally of Part A, which covers inpatient hospital services, and Part B, which covers outpatient services. Beneficiaries may obtain outpatient prescription-drug coverage through a separate Part D plan and may purchase a Medigap policy to reduce cost-sharing exposure. Traditional Medicare generally allows beneficiaries to obtain covered services from any participating provider, but it does not include a comprehensive annual out-of-pocket maximum and does not routinely cover several services that older adults commonly use, including most dental, vision, hearing, and fitness services. (Centers for Medicare & Medicaid Services, 2026b).

Medicare Advantage (MA), also known as Medicare Part C, is a privately administered alternative to Traditional Medicare. Individuals who enroll in MA remain entitled to receive Medicare Part A and Part B benefits, but those benefits are delivered through a private health plan that receives risk-adjusted, capitated payments from the federal government. Most MA plans also incorporate Part D prescription drug coverage. Unlike Traditional Medicare, MA plans commonly require provider networks, utilization-management tools, and plan-specific cost-sharing arrangements. They are also required to establish an annual limit on beneficiaries' out-of-pocket spending for Part A and Part B services. Therefore, the choice between Traditional Medicare and MA involves a tradeoff between relatively broad provider choice under Traditional Medicare and the more managed benefit structure, financial protections, and additional services that MA plans may offer (Centers for Medicare & Medicaid Services, 2026b; Medicare Payment Advisory Commission, 2025).

This payment system gives MA plans an important role in benefit design. Each plan submits a bid reflecting the projected cost of providing Medicare-covered Part A and Part B services (fee-for-service) in a county. The bid is compared with an administratively determined county benchmark. When a plan's bid falls below the benchmark, a portion of the difference is returned to the plan as a rebate. Plans must use these rebate dollars to provide additional value to beneficiaries through lower premiums, reduced cost sharing, enhanced Part D coverage, or supplemental benefits. The amount available for supplemental bene-

fits therefore varies across plans and markets according to plan bids, county benchmarks, quality ratings, and other features of the MA payment system (Medicare Payment Advisory Commission, 2025). This institutional structure makes supplemental benefits a prominent dimension on which MA plans compete for enrollment.

MA has become a central component of the Medicare program. As of 2026, approximately 35.2 million people, representing 55 percent of eligible Medicare beneficiaries, were enrolled in an MA plan (Freed et al., 2026b). The growing scale of MA means that plan decisions about supplemental benefits can affect the healthcare access and preventive behavior of a large share of the older U.S. population. In addition, MA enrollment is voluntary, and individuals may select plans on the basis of premiums, provider networks, prescription-drug coverage, health status and preferences, expected healthcare use, and the perceived value of supplemental benefits. This selection is important when evaluating the effects of MA as beneficiaries who choose MA may be systematically different from those who remain in Traditional Medicare or no Medicare at all.

2.2 Supplemental Benefits in Medicare Advantage

MA plans must cover all medically necessary services covered by Medicare Part A and Part B, but they may also offer benefits beyond the Traditional Medicare package. These supplemental benefits can be grouped into several broad categories. First, plans may reduce beneficiary cost sharing for Medicare-covered services. Second, they may use rebates to reduce Part B or Part D premiums. Third, plans may enhance prescription-drug coverage. Fourth, plans may provide non-Medicare services such as routine dental care, eye examinations and eyeglasses, hearing examinations and hearing aids, meals after hospitalization, transportation to medical appointments, remote access technologies, in-home support, acupuncture, and fitness benefits (Medicare Payment Advisory Commission, 2025; Freed et al., 2026a). Some plans, particularly special needs plans, may also offer benefits intended to address health-related social needs, subject to federal eligibility and targeting requirements.

Fitness coverage is one of the most widespread MA supplemental benefits. Traditional Medicare does not cover a gym membership or a general fitness program, whereas most individual MA plans include these services as part of their supplemental benefit package (Centers for Medicare & Medicaid Services, 2026a). The share of MA enrollees in plans offering a fitness benefit increased from approximately 80 percent in 2018 to 93 percent in

2025 (Medicare Payment Advisory Commission, 2025). Common fitness programs include SilverSneakers, Renew Active, and SilverFit. Although program details differ across plans and locations, these fitness benefits provide free or subsidized access to participating fitness facilities, group exercise classes, online exercise resources, and community-based activities. Some programs also provide exercise opportunities designed for different functional levels, allowing older adults to participate in structured activity.

Fitness benefits may influence physical activity through several mechanisms. The most direct mechanism is a reduction in the financial barrier to exercise. A free or subsidized membership eliminates recurring fees that may otherwise deter participation, particularly among beneficiaries with limited disposable income. The benefit may also reduce information and search costs by directing beneficiaries to participating facilities and organized programs. Group classes can provide instruction, scheduling, and social interaction, while access to multiple facilities may reduce travel and convenience barriers. The inclusion of fitness coverage in a health plan may additionally signal that exercise is a legitimate component of healthcare and healthy aging, increasing the salience of physical activity at the time of Medicare enrollment.

However, benefit availability alone may not produce sustained participation. Survey evidence suggests that actual use of gym and fitness benefits is substantially lower than the share of beneficiaries enrolled in plans that offer them, and administrative data on supplemental-benefit use remain incomplete (Medicare Payment Advisory Commission, 2025). Beneficiaries may be unaware of the benefit, lack a conveniently located participating facility, face transportation or mobility barriers, or enroll but attend infrequently. These gaps between availability, take-up, and sustained use imply that an estimate of the effect of MA enrollment captures an intention-to-treat or reduced-form response to access to the overall MA benefit package rather than the treatment-on-the-treated effect of fitness-program participation.

Fitness benefits may also affect plan selection. Because such benefits are likely to be more valuable to individuals who are already relatively healthy, mobile, or interested in exercise, plans may use them to attract favorable risks. Consistent with this concern, Cooper and Trivedi (2012) find that the introduction of a fitness benefit was associated with enrollment by beneficiaries with more favorable self-reported health. Consequently, comparisons between fitness-program users and nonusers may confound the effect of the program with preexisting differences in health, motivation, and exercise preferences. This selection problem motivates research designs that examine changes generated by plausibly exogenous access rather than

relying only on cross-sectional comparisons of participants and nonparticipants.

2.3 Determinants and Barriers to Exercise

The persistence of physical inactivity despite its well-established health benefits is a central puzzle in health behavior. Exercise produces benefits that accumulate over time, including better cardiovascular and metabolic health, greater functional capacity, and lower risks of disability, premature mortality, and falls later in life. However, its immediate costs are high. Exercise requires time, effort, planning, and direct expenditures for memberships, equipment, and transportation. It may also generate short-run discomfort, embarrassment, fatigue, or fear of injury. Thus, individuals make exercise decisions by trading immediate and relatively certain costs against health benefits that are delayed and difficult to observe.

For older adults, the barriers can be particularly substantial. A systematic review by Franco et al. (2015) identifies physical limitations, pain, fear of falling, chronic conditions, competing priorities, lack of motivation, limited social support, and problems with access and affordability as recurring barriers to physical activity among older adults. Qualitative evidence similarly emphasizes aging-related health limitations, inadequate professional guidance, and the absence of accessible exercise opportunities (Bethancourt et al., 2014). Transportation difficulties, unsafe or poorly maintained neighborhoods, unfavorable weather, limited availability of age-appropriate programs, and the distance to recreational facilities can further increase the effective cost of exercise.

Exercise behavior is also shaped by psychological and social factors. Self-efficacy, perceived competence, enjoyment, prior exercise experience, social support, and beliefs about the benefits and risks of activity are consistently associated with participation (Bauman et al., 2012; Koeneman et al., 2011). Older adults may avoid vigorous physical activity, or even moderate activity, as they are uncertain about which activities are safe or appropriate for their health conditions. Thus, advice from a physician or exercise professional may reduce informational barriers, while supervised or group-based programs may reduce fear and increase confidence. Social interaction can itself be an important benefit of participation, particularly for retired or socially isolated adults. On the other hand, the absence of an exercise partner or a welcoming environment can discourage attendance even when the financial barrier is low.

It is useful to distinguish the decision to initiate exercise from the decision to maintain

it. A subsidy or new insurance benefit may induce an individual to visit a gym or attend a workout class initially, but sustained participation requires that exercise become compatible with daily routines and that its perceived benefits exceed its continuing time and costs. The determinants of maintenance may therefore be different from those of initial take-up (Koene-man et al., 2011). Repeated participation can generate skills, familiarity, social relationships, and habits that may lower the future cost of exercise. Alternatively, participation may fade once the perceived benefit diminishes or when health, caregiving, work, or transportation constraints reemerge. This distinction is directly relevant to MA fitness coverage as a benefit embedded in Medicare Advantage can create an initial opportunity to exercise without necessarily ensuring a persistent behavioral change.

2.4 Behavioral Explanations and Financial Incentives for Exercise

Behavioral economics provides an additional explanation for the gap between intentions and exercise behavior. Individuals may place disproportionate weight on immediate costs relative to delayed health benefits, a pattern commonly described as present bias. A person may plan to exercise in the future but repeatedly postpone the activity when the time to exercise arrives. Limited attention, forgetfulness, uncertainty about future schedules, and inaccurate beliefs about one’s future motivation may reinforce this tendency. Thus, exercise may remain below the level that an individual would choose under full commitment even when the person understands its long-run benefits.

Evidence from health-club contracts highlights these problems. DellaVigna and Malmendier (2006) document that many consumers select monthly gym contracts that are more expensive than paying per visit given their realized attendance and that they delay cancelling memberships they rarely use. These patterns are consistent with overoptimism about future attendance and difficulty following through on exercise intentions. Similarly, Acland and Levy (2015) find that individuals overpredict their future gym attendance and present evidence consistent with partial belief about self-control as well as habit formation. A free gym membership can remove an important financial barrier, but it may not fully address procrastination or overoptimism.

A growing experimental literature examines whether making the reward for exercise more immediate can overcome these barriers. Charness and Gneezy (2009) find that temporary financial incentives substantially increase gym attendance during the incentive period and can produce some persistence after payments end. Acland and Levy (2015) also find the ev-

idence indicating that an intervention can generate post-intervention attendance consistent with habit formation. In contrast, Royer et al. (2015) find that a short-term workplace incentive alone produces relatively small lasting changes, while allowing participants to place their own money at risk through a commitment contract generates more durable effects. Thus, these studies suggest that incentives can be effective at initiating exercise but that persistence depends on whether the intervention changes routines, habits, beliefs, or commitment.

The design of an incentive also matters. An immediate reward tied to each completed activity directly offsets the present effort cost, whereas a general membership subsidy reduces a fixed financial cost regardless of actual attendance. Thus, a membership benefit may generate large gains for individuals for whom the price is the main barrier but smaller effects for those constrained by time, mobility, transportation, or low motivation. In contrast, an in-kind fitness benefit may have advantages over a one-time cash incentive. It can remain available throughout the plan year, repeatedly remind beneficiaries of the exercise opportunity, connect them to facilities and instructors, and support social participation. Taken together, MA fitness benefits combine elements of a price subsidy, information intervention, and access intervention, although they generally do not condition a financial payment directly on the exercise.

However, a reduction in financial barrier to exercise may not affect all populations equally. It can be more binding for beneficiaries with lower income or education, while facility access, neighborhood safety, or health limitations may be more important factors for other groups. Accordingly, the behavioral effect of an MA fitness benefit may depend on whether the benefit addresses the specific barriers faced by beneficiaries and whether a participating exercise option is practically accessible.

3 Data

The main data source we use is the Health and Retirement Study (HRS), a nationally representative longitudinal survey of older adults in the United States conducted biennially by the University of Michigan. We use survey waves from 2012 to 2022 providing detailed information on insurance coverage, demographics, labor market status, and self-reported physical activity. As our empirical strategy exploits the Medicare eligibility threshold at age 65, a key variable in the analysis is continuous age. Thus, we construct a precise age by using the respondent’s birth year and month with the survey year and month. This allows us to

implement the difference-in-discontinuities design with age as the running variable centered at 65.

The primary outcome is vigorous physical activity based on the following HRS question: “*How often do you take part in sports or activities that are vigorous, such as running or jogging, swimming, cycling, aerobics or gym workout, tennis, or digging with a spade or shovel?*”. Respondents report one of five levels (“Every day,” “More than once a week,” “Once a week,” “One to three times a month,” or “Hardly ever or never”). We construct the main outcome as a binary indicator equal to one if the respondent reports vigorous exercise every day, and zero otherwise. This is an intensive margin of physical activity. As shown in Table 1, only about 3 percent of respondents in the analytic sample report daily vigorous exercise. We focus on this threshold for both policy and measurement reasons. First, daily vigorous activity is the HRS category most closely aligned with federal guideline concordant physical activity. The 2018 *Physical Activity Guidelines for Americans* recommend that older adults engage in 75 - 150 minutes per week of vigorous-intensity aerobic activity or 150 - 300 minutes per week of moderate-intensity aerobic activity, along with muscle-strengthening and multicomponent exercise (U.S. Department of Health and Human Services (2018)). As the HRS reports exercise frequency but not minutes, daily vigorous exercise provides the clearest available proxy for respondents who are likely to meet or exceed the federal guideline of vigorous-intensity physical activity.

In addition, daily vigorous exercise corresponds closely to the type of structured, gym-based activity that Medicare Advantage fitness benefits are designed to encourage. The HRS question explicitly includes activities such as running, swimming, cycling, aerobics, gym workouts, and tennis, all of which could be affected by free or subsidized access to fitness facilities. Although daily vigorous exercise is a demanding outcome, this feature strengthens the interpretation of the result that detecting a significant effect on such a low-prevalence and intensive margin suggests that MA-linked fitness benefits may shift some older adults into a substantively meaningful level of preventive health behavior. Thus, we interpret the main estimate as evidence on a high-value, federal guideline-concordant behavioral margin rather than as a measure of all physical activity. In Appendix Table 4A, we additionally examine broader binary cutoffs and a workout-frequency score to assess whether the result is sensitive to the specific daily-exercise threshold.

In the main analysis, we define insurance groups using observed Medicare coverage histories. The MA group consists of respondents who are observed in Medicare Advantage

but never in Traditional Medicare during the sample period, while the comparison group includes respondents observed in Traditional Medicare or never enrolled in Medicare at the time of the survey. To ensure balanced identification on both sides of the eligibility threshold, we restrict the samples to respondents who are observed in at least one wave before and one wave after age 65. Including individuals who are only observed prior to age 65 would contribute observations exclusively to the left side of the cutoff, introducing asymmetry in the local comparison and potentially biasing the discontinuity estimate. This restriction ensures that the DiDC estimate is identified from within-person variation across the eligibility threshold rather than from cross-sectional comparisons between individuals who happen to fall on only different sides of age 65. In addition, we drop the year 2020 as the COVID-19 pandemic substantially disrupted physical activity opportunities and health behaviors, including widespread gym closures that may have directly affected access to fitness services. We separately assess the sensitivity of our results to including 2020 observations in the robustness analysis.

Table 1 reports pre-65 baseline characteristics for the estimation sample. Among respondents in the MA group, 59.8 percent are female, 49.0 percent are non-Hispanic White, and 26.6 percent have a college degree or higher. The corresponding proportions in the Non-MA group are 55.2 percent, 59.6 percent, and 34.1 percent, respectively. Average household income is 77.7 thousand in the MA group and 93.0 thousand in the Non-MA group, while the mean health-condition index is 1.98 and 1.87, respectively. Daily vigorous exercise is relatively uncommon in both groups, with 2.6 percent of respondents in the MA group and 3.4 percent of respondents in the Non-MA group reporting vigorous exercise every day. These statistics provide a descriptive overview of the demographic, socioeconomic, health, and physical activity characteristics of respondents prior to age 65.

4 Empirical Strategy

A central challenge is that both Medicare Advantage and Traditional Medicare enrollment are endogenous choices. Individuals who select MA over TM or no Medicare may differ systematically in their health preferences, socioeconomic status, underlying health conditions, and baseline physical activity. A direct comparison of exercise behavior across insurance groups therefore conflates the behavioral effect associated with MA coverage with selection on unobserved characteristics.

We address this challenge using a difference-in-discontinuities (DiDC) design following Grembi et al. (2016). This method exploits the exogenous timing of Medicare eligibility at age 65 and compares the discontinuity in exercise behavior among individuals in the MA group with the corresponding discontinuity among individuals in the non-MA comparison group. Medicare eligibility produces a sharp increase in Medicare coverage and MA enrollment at age 65. However, other factors that may affect exercise, such as retirement and Social Security claiming, may also change around the same threshold. A standard regression discontinuity design would therefore conflate the MA-specific behavioral response with other age-65 shocks, while a traditional difference-in-difference design would not directly isolate the discontinuous change occurring at the eligibility threshold. The DiDC framework addresses both limitations by differencing the age-65 discontinuity experienced by the MA group from the corresponding discontinuity experienced by the non-MA group. Under the identifying assumption that other age-65 shocks affect both groups similarly, the remaining differential discontinuity captures the MA-specific behavioral response at the eligibility threshold.

Formally, the DiDC estimand is:

$$\begin{aligned} \text{DiDC} = & \left[E(\textit{Workout} \mid \textit{Post65}, MA=1) - E(\textit{Workout} \mid \textit{Pre65}, MA=1) \right] \\ & - \left[E(\textit{Workout} \mid \textit{Post65}, MA=0) - E(\textit{Workout} \mid \textit{Pre65}, MA=0) \right]. \end{aligned} \quad (1)$$

To clarify what this estimand recovers, we decompose the within-group differences for each insurance group as follows :

$$\begin{aligned} \Delta^{MA} &= (\textit{Workout}_{65,\text{post}}^{MA} - \textit{Workout}_{65,\text{pre}}^{MA}) \\ &= \underbrace{\text{Universal 65 shock}}_{\text{common to all}} + \text{MA-specific selection} + \text{MA fitness benefit} + \underbrace{\text{Time-invariant selection}}_{\text{fixed traits}} \\ \Delta^{\text{non-MA}} &= (\textit{Workout}_{65,\text{post}}^{\text{non-MA}} - \textit{Workout}_{65,\text{pre}}^{\text{non-MA}}) \\ &= \underbrace{\text{Universal 65 shock}}_{\text{common to all}} + \text{Non-MA-specific selection} + \underbrace{\text{Time-invariant selection}}_{\text{fixed traits}} \end{aligned}$$

Within-group differencing cancels time-invariant selection, and taking the difference across groups eliminates the universal age-65 shock common to both groups :

$$\text{DiDC} = \Delta^{MA} - \Delta^{\text{non-MA}} = \text{MA fitness benefit} + (\text{MA-specific selection} - \text{Non-MA-specific selection})$$

The DiDC estimand therefore removes confounders common to both groups at age 65 as well as time-invariant individual differences in exercise preferences. However, the remaining threat to identification is the possibility that MA and non-MA enrollees already exhibit different exercise trajectories before age 65. To assess this, we perform an event study to check differential pre-trends in exercise behavior prior to age 65.

4.1 Difference-in-Discontinuity Specification

We estimate the following reduced-form equation as our main specification:

$$\begin{aligned} Y_{it} = & \alpha + \beta (\text{Post}_{it} \times MA_i) + \gamma_1 \text{Post}_{it} + \gamma_2 MA_i \\ & + f(\text{Age}_{it}) + \text{Post}_{it} f(\text{Age}_{it}) + MA_i f(\text{Age}_{it}) \\ & + (\text{Post}_{it} \times MA_i) f(\text{Age}_{it}) + \boldsymbol{\delta}^\top X_{it} + \mu_i + \tau_t + \varepsilon_{it}, \end{aligned} \quad (2)$$

where Y_{it} is a binary indicator equal to 1 if individual i reports daily vigorous exercise at time t , and 0 otherwise. Post_{it} equals 1 if $\text{Age}_{it} \geq 65$ and 0 otherwise. MA_i equals 1 for individuals ever observed in Medicare Advantage during the study period and 0 for individuals in the non-MA comparison group. $f(\text{Age}_{it})$ is a local-linear function of age, estimated separately on each side of the cutoff and allowed to differ by MA status. X_{it} includes time-varying controls including health conditions, marital status, co-residence with children, and total household income. Health conditions are the sum of indicators for whether a doctor has ever told respondents that they had any of the conditions including high blood pressure, diabetes, cancer, lung disease, heart disease, stroke, psychiatric problems and arthritis. μ_i and τ_t denote individual and year fixed effects, respectively, which absorb time-invariant unobservables and common aggregate shocks. Standard errors are clustered at the individual level.

The coefficient of interest is β , which captures the differential discontinuity at age 65 in the probability of daily vigorous exercise for MA enrollees relative to non-MA individuals. Under the identifying assumption that both groups would have experienced similar discontinuities in exercise at age 65 absent MA-specific fitness benefits, β recovers the reduced-form causal effect of MA enrollment on exercise behavior at the eligibility threshold. The local-linear age controls on each side of the cutoff, interacted with both Post_{it} and MA_i , flexibly absorb smooth age trends in exercise that may differ across groups, ensuring that β is identified from the discontinuous jump at the threshold rather than extrapolation from age trends.

We select the estimation bandwidth using the data-driven procedure of Calonico et al. (2014), which selects the bandwidth by minimizing the asymptotic mean squared error of the local polynomial estimator. This approach balances bias and variance in estimation near the threshold without relying on an artificial bandwidth choice. The resulting data-driven bandwidth is approximately 3.5 years. To assess sensitivity, we report estimates across a range of alternative bandwidths and present the full bandwidth sensitivity plot.

To assess whether the two groups exhibit detectable differential trajectories before age 65 and to describe how the MA–non-MA difference evolves after eligibility, we estimate an event-study specification. Let b_{it} index one-year age bins relative to age 65, so that $b_{it} = 0$ corresponds to the eligibility year and $b_{it} = -1$ serves as the omitted reference category which is a year before Medicare eligibility (age 64) :

$$\begin{aligned}
Y_{it} = & \alpha + \sum_{k \neq -1} \beta_k \mathbb{1}\{b_{it} = k\} MA_i + \sum_{k \neq -1} \phi_k \mathbb{1}\{b_{it} = k\} \\
& + g(Age_{it}) + Post_{it} g(Age_{it}) + MA_i g(Age_{it}) + (Post_{it} \times MA_i) g(Age_{it}) \\
& + \delta' X_{it} + \mu_i + \tau_t + \varepsilon_{it}.
\end{aligned} \tag{3}$$

The coefficients β_k measure the age-bin-specific difference in the probability of daily vigorous exercise between the MA and non-MA groups, relative to the corresponding difference in the omitted year immediately before Medicare eligibility (age 64). The pre-65 coefficients are used to assess whether there are detectable differential exercise trajectories before age 65 and the post-65 coefficients describe how the MA–non-MA difference evolves after the eligibility threshold.

5 Results

5.1 Discontinuities in Workout Behavior

We begin by presenting non-parametric evidence on exercise behavior around the Medicare eligibility threshold. Figure 1A and Figure 1B plot three-month binned means of probability of daily vigorous exercise, with linear fits estimated on each side of age 65 separately for individuals in the Medicare Advantage group and those in the non-MA comparison group. Prior to age 65, the probability of daily vigorous exercise is lower in the MA group than non-MA group but, exactly at the eligibility threshold, MA enrollees exhibit a discrete increase

in daily vigorous exercise probability, whereas non-MA enrollees do not. This visual jump anticipates the parametric DiDC estimates and suggests that MA-linked fitness benefits may trigger behavioral responses.

Figure 1 also illustrates why simple comparisons around age 65 may be insufficient. A variety of health and lifestyle transitions including retirement, changes in caregiving responsibilities, and broader age-related changes in physical activity may occur near the Medicare eligibility threshold. These potentially concurrent changes motivate the DiDC design, which differences the age-65 discontinuity for the MA group from the corresponding discontinuity for the non-MA group.

5.2 Difference-in-Discontinuities Results

Figure 2 presents the first-stage discontinuity in Medicare Advantage enrollment at the Medicare eligibility cutoff. Consistent with the institutional setting, Medicare Advantage enrollment increases sharply at age 65, indicating that Medicare eligibility generates substantial variation in MA enrollment. This pattern supports the empirical relevance of the age-65 cutoff for the DiDC design.

Table 2 reports reduced-form DiDC estimates of the differential age-65 discontinuity in the probability of daily vigorous exercise for the MA group relative to the non-MA group across several bandwidth choices. Our preferred specification uses the MSE-optimal bandwidth following Calonico et al. (2014). In our setting, the selected reference bandwidth is approximately 3.5 years, allowing the local estimation window to be determined by the statistical properties of the data rather than by an arbitrary bandwidth choice.

At this data-driven bandwidth, the estimated differential discontinuity is 0.039, corresponding to a 3.9 percentage point increase in the probability of daily vigorous exercise for the MA group relative to the corresponding age-65 change in the non-MA group. This estimate should not be interpreted as the difference between the raw pre- and post-65 means reported in Table 2. Those means average all observations within the selected bandwidth, whereas the regression estimates the discontinuous change at the age-65 threshold after accounting for group-specific local age trends, individual fixed effects, year fixed effects, and observed covariates. The estimate therefore implies approximately four additional daily exercisers per 100 MA individuals at the threshold, relative to the counterfactual change represented by the non-MA group.

To assess sensitivity to the choice of local window, Panel B of Table 2 reports estimates using narrower and wider bandwidths around the cutoff, and Figure 3 plots the corresponding point estimates and 95% confidence intervals across bandwidths. The estimates are noisier at very narrow bandwidths, reflecting limited precision in a small local sample, but remain positive and broadly stable across bandwidths between 2.5 years and 4.25 years. This pattern indicates that the main result is not driven by a single bandwidth choice.

As an additional diagnostic to check differential pre-trends, Appendix Figure A1 reports event-study estimates around age 65. The pre-65 coefficients are statistically insignificant, providing no evidence of differential exercise trends between the MA and non-MA groups before Medicare eligibility. At age 65, the estimated MA–non-MA differential rises by approximately 3.5 percentage points. The estimates in subsequent age bins are generally smaller and imprecisely estimated. This pattern descriptively highlights the attenuation following the initial age-65 response rather than a clearly sustained effect. However, as enrolling in MA or TM are both endogenous choices, the post-65 coefficients are not interpreted as causal dynamic treatment effects. Instead, they describe how the conditional MA–non-MA exercise differential evolves after Medicare eligibility.

6 Heterogeneity Analysis

We examine whether the reduced-form association between MA enrollment and daily vigorous exercise differs across demographic subgroups in three dimensions including Gender, Race/Ethnicity (White and non-White), Educational attainment (College degree and No College degree). For each subgroup analysis, we estimate the same reduced-form DiDC specification used in the main analysis and report the DiDC estimates using data-driven bandwidths following Calonico et al. (2014).

The heterogeneity results presented in Figure 4 underscore the meaningful variations in behavioral responses to Medicare Advantage (MA) fitness benefits at age 65. Female beneficiaries experience an increase of approximately 5 percentage points in the probability of exercising daily, while the corresponding estimate for males is smaller and statistically insignificant. The difference by race/ethnicity is particularly pronounced as Non-White beneficiaries experience a large increase of approximately 8 percentage points. We do not detect any significant impact among White beneficiaries. Educational differences mirror this pattern as well. College-educated respondents show small and statistically insignificant effects,

while non-college educated seniors show statistically significant increases of roughly 5 percentage points. Overall, the estimates are larger among women, non-White beneficiaries, and beneficiaries without a college degree.

These patterns suggest that the estimated behavioral response to MA enrollment may be larger among populations that face greater barriers to physical activity. However, the subgroup estimates should not be interpreted as direct evidence that MA fitness benefits reduce disparities as the analysis neither observes fitness-benefit uptake nor directly estimates changes in inequality. Instead, the distribution of the estimates is consistent with the possibility that MA-related changes are more prominent among groups facing greater financial, informational, or access barriers.

7 Robustness Checks

To assess the validity of our results, we perform a comprehensive set of robustness checks that address potential concerns about our main findings. These robustness checks assess whether the estimated differential discontinuity at age 65 is sensitive to model specification, local age-related dynamics, irregularities near the cutoff, alternative outcome definitions, or the exclusion of COVID-era observations.

7.1 Alternative Outcomes

We estimate the same DiDC specification using several alternative outcomes that may contemporaneously change around Medicare eligibility and potentially affect workout frequencies including retirement, private insurance coverage, moderate physical activity, and changes in health conditions. The purpose of these robustness checks is to rule out the possibility that the estimated physical activity jump reflects broader life-cycle transitions around age 65. Consistent with expectations, retirement does not show a meaningful differential discontinuity at age 65, as shown in Table 3. In addition, we find a statistically significant decline in the number of private insurance plans at age 65 for the MA group relative to the non-MA group, which indicates that MA beneficiaries are not simultaneously holding private coverage that could independently encourage exercise. The decline in private coverage suggests that changes in other private insurance coverage are unlikely to account for the estimated differential in vigorous exercise.

To further assess whether the estimated MA effect reflects an increase in overall physical activity rather than a mechanism linked to structured fitness benefits, we examine a separate outcome measuring daily engagement in moderate physical activities. We use HRS question, *"How often do you take part in sports or activities that are moderately energetic such as, gardening, cleaning the car, walking at a moderate pace, dancing, floor or stretching exercise"*. Table 3 shows that the differential discontinuity at age 65 in moderate physical activity is small and statistically insignificant. The absence of a significant effect on daily moderate activity suggests that our primary findings on vigorous physical activity are unlikely to be driven by broad shifts in activity or general health motivation triggered by Medicare eligibility. Instead, the results are consistent with a more targeted mechanism where enrollment in MA with fitness benefits primarily affect structured or vigorous exercise behaviors, such as gym-based workouts, rather than everyday moderate physical activity.

As an additional diagnostic, Appendix Figure A2 examines the discontinuity of several observable characteristics around age 65, including health conditions, marital or partnership status, co-residence with children, and household income. Most of these characteristics appear relatively smooth around the eligibility threshold, although the health condition measure shows some visual evidence of a change for the MA group. We therefore further examine whether health conditions exhibit a differential discontinuity between the MA and non-MA groups. As reported in column (4) of Table 3, the estimated differential discontinuity in health conditions is small and statistically insignificant, providing no evidence of a differential change in health conditions at age 65 between the two groups.

We also examine whether the main result extends to alternative measures of vigorous physical activity. Columns (1)–(3) of Appendix Table 4A uses progressively broader binary definitions of vigorous exercise such as exercising more than once per week, exercising at least once per week, and engaging in any vigorous exercise. The estimated differential discontinuities for these outcomes are small and statistically insignificant. These results indicate that the main finding does not extend broadly to less frequent definitions of vigorous exercise and is instead concentrated at the daily exercise margin.

Column (4) of Appendix Table 4A uses an alternative workout-frequency score that retains some information from the original survey categories. For the most parsimonious analysis, the outcome measure is coded as 7 for respondents who report vigorous activity every day, 1 for those who report activity more than once per week, once per week, or one to three times per month, and 0 for those who report no vigorous activity. The estimated differential

discontinuity is 0.178 and is statistically significant at the 10 percent level. As this outcome is a frequency score rather than a binary indicator, the coefficient is interpreted in score units rather than percentage points. Taken together, the results indicate that the MA-associated increase in vigorous exercise is concentrated primarily at the daily exercise margin, with more limited evidence of an increase in overall vigorous exercise frequency.

7.2 Placebo Cutoffs and Sample Specifications

We conduct three additional robustness checks to assess whether the main DiDC estimate is driven by local irregularities around the age-65 cutoff, by spurious age-related discontinuities, or by the exclusion of COVID-era observations. First, we estimate a donut-hole specification that excludes observations within one month on either side of age 65. This test addresses concerns that local administrative errors, reporting issues, or mechanical changes around the Medicare eligibility threshold may drive the results. Second, we estimate a placebo DiDC specification using age 64 and 66 as placebo cutoffs. As age 64 and 66 do not correspond to the main Medicare eligibility threshold, we should not observe a meaningful differential discontinuity at these ages. Third, we re-estimate the main specification including year 2020 (COVID) observations that are excluded from the main specification.

Table 4 reports the results using the reference bandwidth for each robustness exercise. The donut-hole estimate remains positive and similar in magnitude to the main estimate, suggesting that the result is not driven by administrative or reporting errors. The placebo estimates at age 64 and 66 are both small and statistically insignificant, which supports the interpretation that the main discontinuity is specific to the true Medicare eligibility threshold. Finally, when year 2020 (COVID) observations are included, the estimate still remains positive and significant but is smaller in magnitude. This finding highlights that the main result is not mechanically driven by excluding the COVID period, during which nationwide lockdown policies and gym closures significantly disrupted access to structured exercise across both MA and non-MA groups.

7.3 Heterogeneity by Fitness Benefit Availability over Time

The proportion of Medicare Advantage which covers fitness benefits has expanded substantially during the late 2010s and early 2020s (Freed et al., 2024). They report that in 2022, 94% of individual Medicare Advantage plans offer a fitness benefit, underscoring the nearly universal availability of supplemental fitness coverage in recent years. We assess whether the estimated effect of MA enrollment is larger in later years, when fitness benefits were more

widely available

For this analysis, we split the sample into an early period (2012 - 2014) and a later period (2016 - 2022). The data-driven bandwidth following Calonico et al. (2014) for the 2012 - 2014 period is 1.47 years, which yields only 630 observations and therefore substantially reduces precision. To facilitate comparison across periods, we use a bandwidth of 4.100 years, corresponding to the bandwidth used for the 2016 - 2022 period. As shown in Table 5, the estimated differential discontinuity is small and statistically insignificant in 2012 - 2014, whereas the estimate is larger and statistically significant in 2016 - 2022. These results are consistent with stronger behavioral responses in the later period, when fitness benefits were more broadly available in Medicare Advantage plans.

8 Conclusion

This paper studies whether insurance benefit design can change preventive health behavior at older ages. Using nationally representative panel data from the Health and Retirement Study from 2012 to 2022, we exploit the Medicare eligibility threshold at age 65 as a source of exogenous variation and estimate a difference-in-discontinuities design comparing MA enrollees with non-MA individuals. The design differences out age-65 shocks common to both groups, including retirement, Social Security claiming, and broader age-related changes in health behavior, while isolating the MA-specific discontinuity in daily vigorous exercise.

The results show that MA enrollment increases the probability of daily vigorous exercise at Medicare eligibility. The estimated effect is meaningful given the low baseline prevalence of daily vigorous activity among older adults. Although daily vigorous exercise is an intensive margin and does not directly measure total weekly minutes of activity, it captures a demanding, guideline-concordant preventive health behavior that is closely aligned with the structured exercise opportunities offered by MA fitness benefits. The subgroup estimates are larger among women, non-White beneficiaries, and adults without a college degree. While the analysis does not directly measure reductions in inequality, the concentration of effects among these groups suggests that supplemental fitness benefits may help reduce disparities in preventive health behaviors at later life.

As Medicare Advantage continues to expand, supplemental benefits are likely to play an increasingly important role in shaping the non-medical dimensions of health insurance. The findings in this paper suggest that benefit design can induce preventive health behav-

ior among older adults. Determining whether these behavioral responses persist beyond the initial Medicare eligibility transition will require research designs capable of identifying longer-term effects.

References

- Acland, D. and Levy, M. R. (2015). Naivet  , projection bias, and habit formation in gym attendance. *Management Science*, 61(1):146–160.
- Bauman, A. E., Reis, R. S., Sallis, J. F., Wells, J. C., Loos, R. J. F., Martin, B. W., and Lancet Physical Activity Series Working Group (2012). Correlates of physical activity: Why are some people physically active and others not? *The Lancet*, 380(9838):258–271.
- Belza, B., Shumway-Cook, A., Phelan, E. A., Williams, B., Snyder, S. J., and LoGerfo, J. P. (2006). The effects of a community-based exercise program on function and health in older adults: The EnhanceFitness program. *Journal of Applied Gerontology*, 25(4):291–306.
- Bethancourt, H. J., Rosenberg, D. E., Beatty, T., and Arterburn, D. E. (2014). Barriers to and facilitators of physical activity program use among older adults. *Clinical Medicine & Research*, 12(1–2):10–20.
- Brady, S., D’Ambrosio, L. A., Felts, A., Rula, E. Y., Kell, K. P., and Coughlin, J. F. (2020). Reducing isolation and loneliness through membership in a fitness program for older adults: Implications for health. *Journal of Applied Gerontology*, 39(3):301–310. First published online November 4, 2018.
- Calonico, S., Cattaneo, M. D., and Titiunik, R. (2014). Robust nonparametric confidence intervals for regression-discontinuity designs. *Econometrica*, 82(6):2295–2326.
- Centers for Disease Control and Prevention (2024). Nutrition, physical activity, and weight status. Updated June 3, 2024; accessed August 6, 2026.
- Centers for Medicare & Medicaid Services (2026a). Gym memberships and fitness programs. Medicare.gov. Accessed August 6, 2026.
- Centers for Medicare & Medicaid Services (2026b). Understanding Medicare Advantage plans. CMS Product No. 12026. Accessed August 6, 2026.
- Charness, G. and Gneezy, U. (2009). Incentives to exercise. *Econometrica*, 77(3):909–931.
- Cooper, A. L. and Trivedi, A. N. (2012). Fitness memberships and favorable selection in Medicare Advantage plans. *New England Journal of Medicine*, 366(2):150–157.
- DellaVigna, S. and Malmendier, U. (2006). Paying not to go to the gym. *American Economic Review*, 96(3):694–719.

- Elgaddal, N. and Kramarow, E. A. (2024). Characteristics of older adults who met federal physical activity guidelines for americans: United states, 2022. National Health Statistics Reports 215, National Center for Health Statistics, Hyattsville, MD. Published November 26, 2024; accessed August 6, 2026.
- Franco, M. R., Tong, A., Howard, K., Sherrington, C., Ferreira, P. H., Pinto, R. Z., and Ferreira, M. L. (2015). Older people’s perspectives on participation in physical activity: A systematic review and thematic synthesis of qualitative literature. *British Journal of Sports Medicine*, 49(19):1268–1276.
- Freed, M., Fuglesten Biniek, J., Damico, A., and Neuman, T. (2024). Medicare Advantage 2025 spotlight: A first look at plan premiums and benefits. Published November 15, 2024; updated November 25, 2024; accessed August 6, 2026.
- Freed, M., Fuglesten Biniek, J., Damico, A., and Neuman, T. (2026a). Medicare Advantage in 2026: Premiums, out-of-pocket limits, supplemental benefits, and prior authorization. Published June 5, 2026; accessed August 6, 2026.
- Freed, M., Fuglesten Biniek, J., Damico, A., Ochieng, N., and Neuman, T. (2026b). Medicare Advantage in 2026: Enrollment update and key trends. Published June 5, 2026; updated July 1, 2026; accessed August 6, 2026.
- Grembi, V., Nannicini, T., and Troiano, U. (2016). Do fiscal rules matter? *American Economic Journal: Applied Economics*, 8(3):1–30.
- Hamar, B., Coberley, C. R., Pope, J. E., and Rula, E. Y. (2013). Impact of a senior fitness program on measures of physical and emotional health and functioning. *Population Health Management*, 16(6):364–372.
- Koeneman, M. A., Verheijden, M. W., Chinapaw, M. J. M., and Hopman-Rock, M. (2011). Determinants of physical activity and exercise in healthy older adults: A systematic review. *International Journal of Behavioral Nutrition and Physical Activity*, 8:142.
- Lee, I.-M., Shiroma, E. J., Lobelo, F., Puska, P., Blair, S. N., Katzmarzyk, P. T., and Lancet Physical Activity Series Working Group (2012). Effect of physical inactivity on major non-communicable diseases worldwide: An analysis of burden of disease and life expectancy. *The Lancet*, 380(9838):219–229.
- Medicare Payment Advisory Commission (2025). Supplemental benefits in Medicare Advantage. Report to the Congress: Medicare and the Health Care Delivery System Chapter 2, Medicare Payment Advisory Commission, Washington, DC. Accessed August 6, 2026.

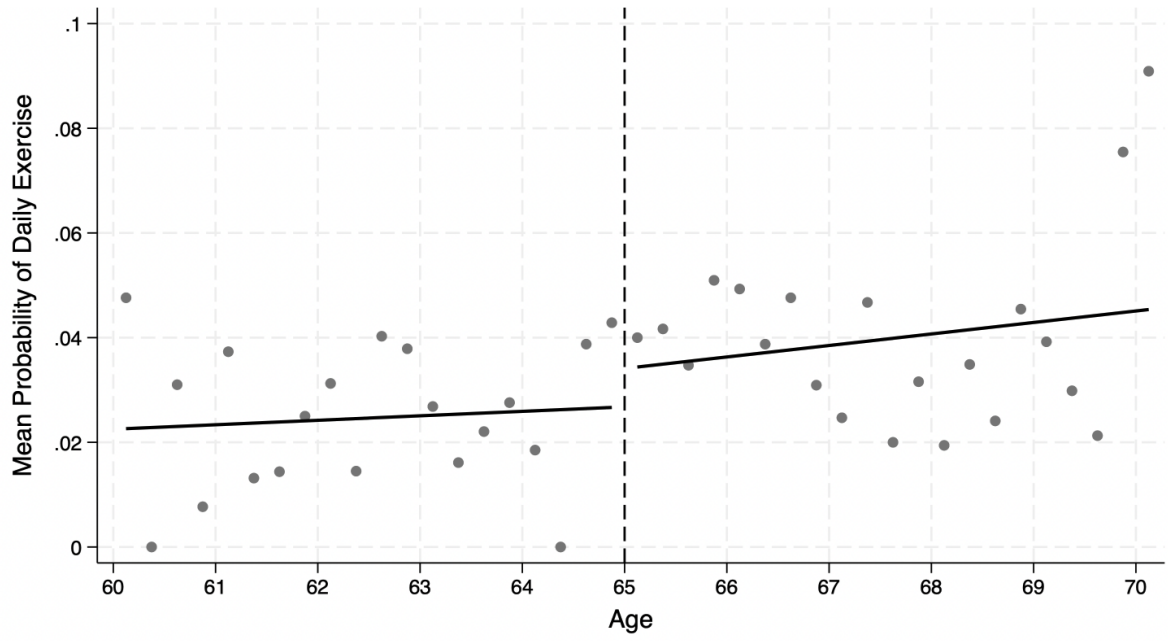
- Nguyen, H. Q., Maciejewski, M. L., Gao, S., Lin, E., Williams, B., and LoGerfo, J. P. (2008). Health care use and costs associated with use of a health club membership benefit in older adults with diabetes. *Diabetes Care*, 31(8):1562–1567.
- Ochieng, N., Freed, M., Fuglesten Biniek, J., Damico, A., and Neuman, T. (2025). Medicare Advantage in 2025: Enrollment update and key trends. Published July 28, 2025; accessed August 6, 2026.
- Royer, H., Stehr, M., and Sydnor, J. (2015). Incentives, commitments, and habit formation in exercise: Evidence from a field experiment with workers at a Fortune-500 company. *American Economic Journal: Applied Economics*, 7(3):51–84.
- U.S. Department of Health and Human Services (2018). Physical activity guidelines for americans. Accessed August 6, 2026.
- Warburton, D. E. R., Nicol, C. W., and Bredin, S. S. D. (2006). Health benefits of physical activity: The evidence. *Canadian Medical Association Journal*, 174(6):801–809.

9 Tables and Figures

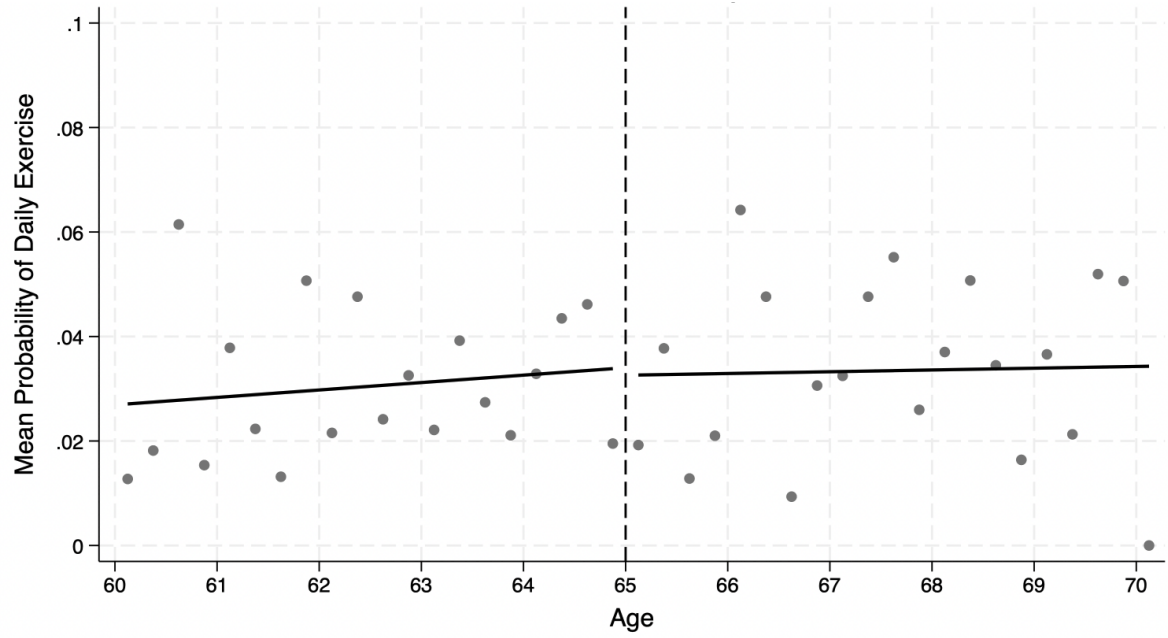
Table 1: Pre-65 Baseline Characteristics by Eventual Medicare Advantage Status

	Non-MA N(%)	MA N(%)	Total N(%)
<i>Gender</i>			
Male	1,016 (44.80)	590 (40.19)	1,606 (42.99)
Female	1,252 (55.20)	878 (59.81)	2,130 (57.01)
<i>Race/Ethnicity</i>			
Non-Hispanic White	1,349 (59.61)	718 (49.04)	2,067 (55.46)
Non-Hispanic Black	416 (18.38)	403 (27.53)	819 (21.97)
Hispanic	406 (17.94)	291 (19.88)	697 (18.70)
Other	92 (4.07)	52 (3.55)	144 (3.86)
<i>Education</i>			
No college degree	1,484 (65.93)	1,068 (73.40)	2,552 (68.86)
College degree or higher	767 (34.07)	387 (26.60)	1,154 (31.14)
<i>Household Characteristics</i>			
Married or partnered	1,602 (70.70)	958 (65.35)	2,560 (68.60)
Living with children	841 (37.08)	560 (38.15)	1,401 (37.50)
Household income (\$1,000s), mean (SD)	92.98 (135.32)	77.71 (131.12)	86.98 (133.87)
<i>Medicare Status</i>			
Enrolled in Medicare	166 (7.32)	150 (10.22)	316 (8.46)
<i>Health Conditions</i>			
Health condition index, mean (SD)	1.87 (1.42)	1.98 (1.45)	1.91 (1.43)
<i>Physical Activity (Vigorous)</i>			
Every day	77 (3.40)	38 (2.60)	115 (3.09)
>1 per week	606 (26.79)	370 (25.27)	976 (26.19)
1 per week	261 (11.54)	165 (11.27)	426 (11.43)
1–3 per month	252 (11.14)	175 (11.95)	427 (11.46)
Never	1,066 (47.13)	716 (48.91)	1,782 (47.83)
Total respondents	2,268	1,468	3,736

Notes: Counts are shown with column percentages in parentheses, except for household income and the health condition index, which are reported as means and standard deviations. Baseline is defined as the respondent's closest available interview before age 65, restricted to interviews conducted from age 60 to less than age 65. Each respondent contributes one observation. The MA group consists of respondents classified as MA-only, while the Non-MA group combines respondents classified as traditional-Medicare-only and respondents in the no-Medicare control group. Race and ethnicity are combined into mutually exclusive categories, with Hispanic respondents classified as Hispanic regardless of race. Married or partnered includes respondents who are married, married with an absent spouse, or partnered. Living with children indicates that at least one child resides in the household. Household income is reported in thousands of dollars. Medicare status indicates whether the respondent was enrolled in Medicare at the baseline interview. The health condition index is based on RAND HRS variable RWCONDE and is defined as the sum of indicators for whether a doctor has ever told the respondent that they had any of the following eight conditions: high blood pressure, diabetes, cancer, lung disease, heart disease, stroke, psychiatric problems, and arthritis. The analytic sample includes 3,736 respondents. Sample sizes vary slightly across characteristics due to missing values.



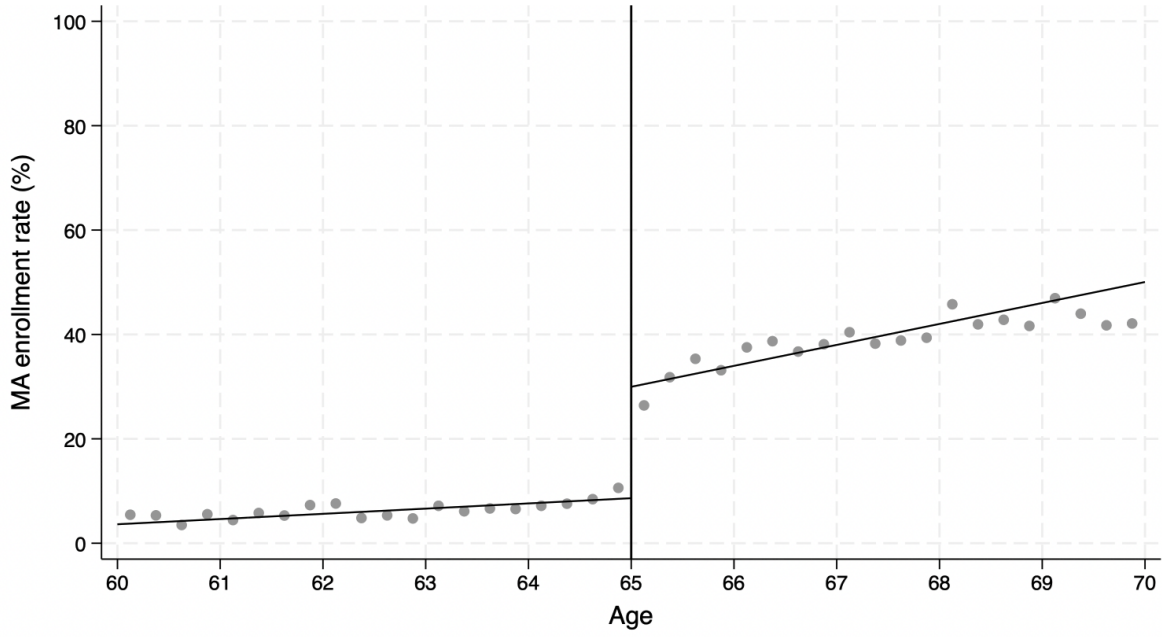
(A) MA-only group



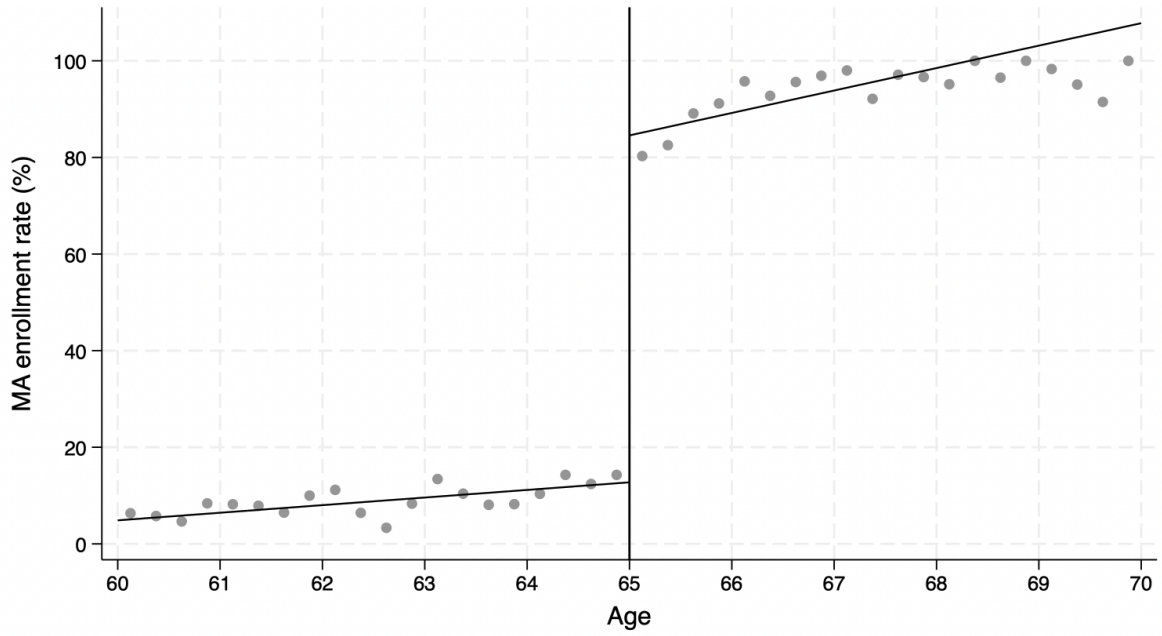
(B) Non-MA group

Figure 1

Figure 1: Dots show 3 months binned means of the probability of daily vigorous exercise. Solid lines show linear fits estimated separately on each side of age 65. The vertical dashed line indicates the age-65 threshold. Panel A presents respondents classified as MA-only, and Panel B presents respondents classified as Non-MA.



(A) Full sample



(B) MA-only group

Figure 2

Figure 2: Dots show 3 months binned means of Medicare Advantage enrollment rates. Solid lines show linear fits estimated separately on each side of age 65. The vertical solid line indicates the age 65 threshold. Panel A presents the full analytic sample, and Panel B presents respondents classified as MA-only.

Table 2: DiDC Estimates of the Differential Jump in Daily Exercise at Age 65

	Panel A: Main	Panel B: Alternative Bandwidths			
	(1) $b = 3.53$	(2) $b = 2$	(3) $b = 3$	(4) $b = 4$	(5) $b = 5$
$\Delta^{MA} - \Delta^{\text{Non-MA}}$ ^a	0.039** (0.015)	0.034 (0.030)	0.046*** (0.017)	0.030** (0.015)	0.019 (0.012)
Pre-65 mean, MA group ^b	0.026	0.022	0.027	0.025	0.025
Post-65 mean, MA group ^b	0.037	0.039	0.039	0.038	0.038
Pre-65 mean, Non-MA group ^b	0.031	0.026	0.031	0.031	0.030
Post-65 mean, Non-MA group ^b	0.033	0.025	0.031	0.033	0.033
Observations	8,706	3,816	7,213	9,995	11,876
R^2	0.499	0.575	0.538	0.477	0.433
Individual fixed effects	✓	✓	✓	✓	✓
Year fixed effects	✓	✓	✓	✓	✓

Notes: Table reports reduced-form DiDC estimates of the differential discontinuity at age 65 in a binary indicator for daily exercise, using different bandwidth choices in years. The dependent variable equals 1 if the respondent reports daily exercise and 0 otherwise. The coefficient $\Delta^{MA} - \Delta^{\text{Non-MA}}$ captures the difference between the jump at age 65 for individuals in the MA group and the corresponding jump for the Non-MA comparison group. For example, the estimate of 0.039 in column (1) corresponds to a 3.9-percentage-point differential jump. The one-year bandwidth is not reported because the HRS is biennial and this window leaves a small and uneven estimation sample around the cutoff. ^a Coefficients are reported in proportion units; multiplying by 100 gives percentage-point effects. ^b Means are computed within the estimation sample for each bandwidth, separately by MA status. Standard errors clustered at the individual level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

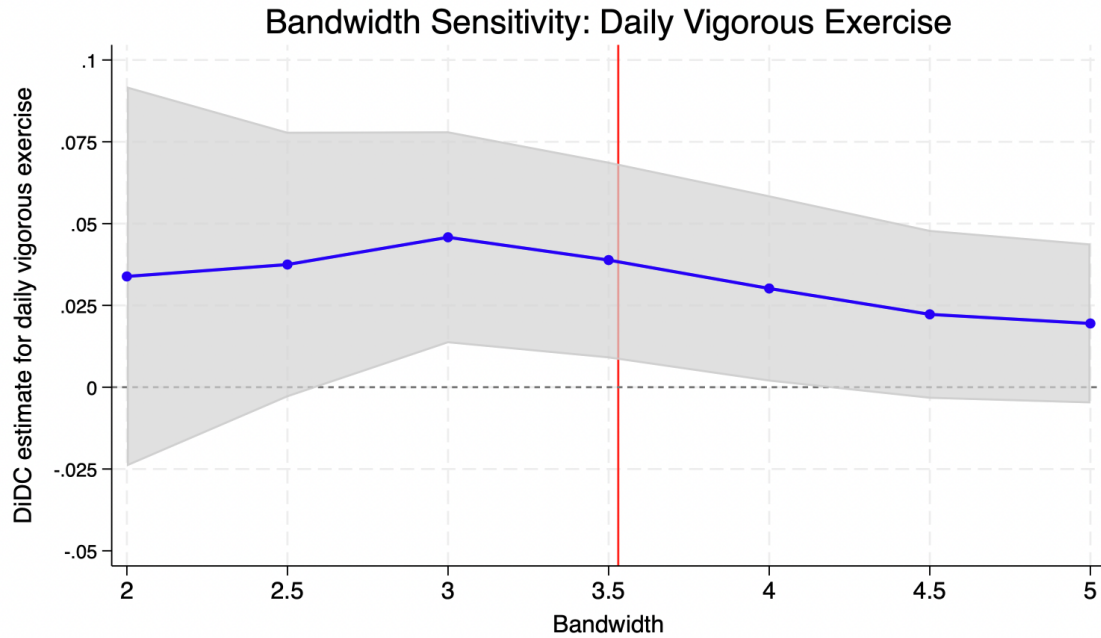


Figure 3: Bandwidth Sensitivity of the Difference-in-Discontinuity Estimate. The figure plots the estimated DiDC coefficient for daily vigorous exercise across alternative bandwidths around the age-65 cutoff. The solid line denotes the point estimate, and the shaded area denotes the 95% confidence interval. The vertical line marks the preferred data-driven bandwidth of 3.53 years.

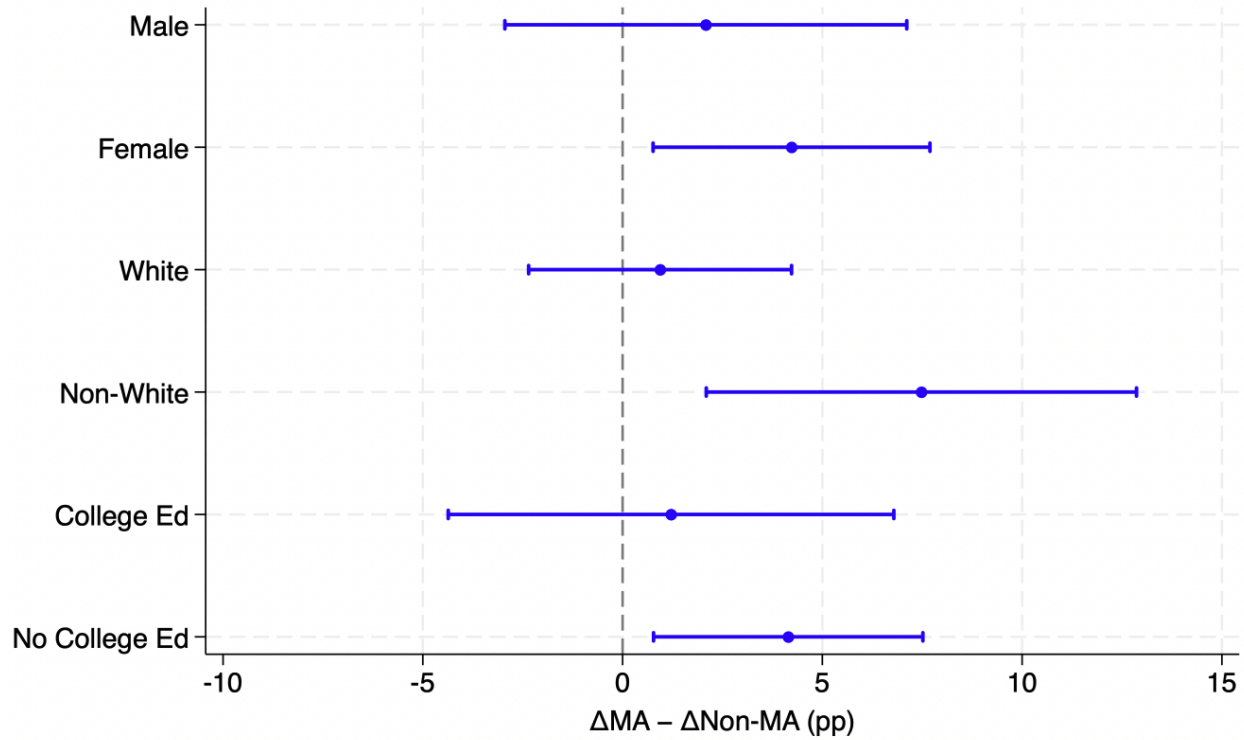


Figure 4: *Notes:* Reduced-form regression discontinuity in difference-in-discontinuities (DiDC) estimates of the differential jump at age 65 in daily exercise across demographic subgroups. Points denote subgroup-specific estimates of $\Delta^{MA} - \Delta^{\text{Non-MA}}$ expressed in percentage points. Horizontal bars represent 95% confidence intervals based on standard errors clustered at the individual level. Estimates are computed using the MSE-optimal bandwidth and include individual and year fixed effects.

Table 3: DiDC Estimates of Differential Jumps at Age 65 Across Outcomes

	Alternative Outcomes			
	Retirement	Private plan	Moderate PA	Health condition
	(1) $b = 3.530$	(2) $b = 3.530$	(3) $b = 4.541$	(4) $b = 3.530$
$\Delta^{MA} - \Delta^{\text{Non-MAa}}$	0.042 (0.029)	-0.382*** (0.043)	0.019 (0.021)	0.027 (0.031)
Mean dependent variable	0.617	0.634	0.096	2.006
Pre-65 mean, MA group ^b	0.565	0.704	0.070	1.949
Post-65 mean, MA group ^b	0.819	0.181	0.099	2.244
Pre-65 mean, Non-MA group ^b	0.499	0.772	0.090	1.818
Post-65 mean, Non-MA group ^b	0.678	0.687	0.122	2.139
Observations	8,544	8,703	11,045	8,747
R^2	0.755	0.689	0.456	0.956
Individual fixed effects	✓	✓	✓	✓
Year fixed effects	✓	✓	✓	✓

Notes: This table reports difference-in-discontinuities (DiDC) estimates of the differential jump at age 65 for several alternative outcomes. Each column uses the preferred bandwidth selected for the corresponding outcome. The dependent variables are retirement status, private plan coverage, daily moderate physical activity, and health condition. The coefficient $\Delta^{MA} - \Delta^{\text{Non-MA}}$ reports the differential discontinuity for the MA group relative to the non-MA group at age 65. Pre- and post-65 means are computed within the estimation sample by MA status and age group. Standard errors clustered at the individual level are reported in parentheses. All specifications include individual and year fixed effects. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Placebo Cutoffs and Sample Specifications

	Donut hole	Placebo cutoffs (64, 66)		Including year 2020
	(1) $b = 3.53$	(2) $b = 4.06$	(3) $b = 4.749$	(4) $b = 5.00$
$\Delta^{MA} - \Delta^{\text{Non-MA}}$	0.037** (0.016)	0.009 (0.013)	-0.009 (0.014)	0.019* (0.011)
Pre-cutoff mean, MA group	0.026	0.025	0.027	0.023
Post-cutoff mean, MA group	0.037	0.035	0.036	0.037
Pre-cutoff mean, Non-MA group	0.031	0.028	0.028	0.030
Post-cutoff mean, Non-MA group	0.034	0.033	0.038	0.034
Observations	8,599	10,447	10,369	13,462
Adjusted R^2	0.186	0.468	0.458	0.405
Individual fixed effects	✓	✓	✓	✓
Year fixed effects	✓	✓	✓	✓

Notes: The table reports reduced-form DiDC robustness estimates for a binary indicator equal to one if the respondent reports exercising every day. Column (1) excludes observations within one month of age 65 and controls for health conditions, household income, living with children, and marital status. Columns (2) and (3) use placebo cutoffs at ages 64 and 66, respectively. Column (4) includes observations from 2020. Means are computed within each estimation sample. Standard errors clustered at the individual level are reported in parentheses. All specifications include individual and year fixed effects. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: DiDC Estimates of the Differential Jump in Daily Exercise at Age 65 by Period

	Dependent Variable: Daily Exercise	
	2012–2014 (1) $b = 4.100$	2016–2022 (2) $b = 4.100$
$\Delta^{MA} - \Delta^{\text{Non-MA}}$	0.016 (0.033)	0.038* (0.023)
Pre-65 mean, MA group	0.014	0.029
Post-65 mean, MA group	0.029	0.042
Pre-65 mean, Non-MA group	0.020	0.034
Post-65 mean, Non-MA group	0.022	0.037
Observations	2,576	6,045
R^2	0.533	0.538
Individual fixed effects	✓	✓
Year fixed effects	✓	✓

Notes: Columns report reduced-form DiDC estimates using a bandwidth of 4.100 years around age 65. For 2012–2014, the data-driven bandwidth was 1.47 years with only 630 observations. We use the 4.100-year bandwidth to match the specification used for the 2016–2022 period and improve comparability across periods. The coefficient $\Delta^{MA} - \Delta^{\text{Non-MA}}$ captures the difference between the jump at age 65 for individuals in the MA group and the corresponding jump for the non-MA comparison group. Pre- and post-65 means are calculated within the corresponding estimation sample. Standard errors clustered at the individual level are reported in parentheses. All specifications include individual and year fixed effects. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A Appendix- Additional Figures and Tables

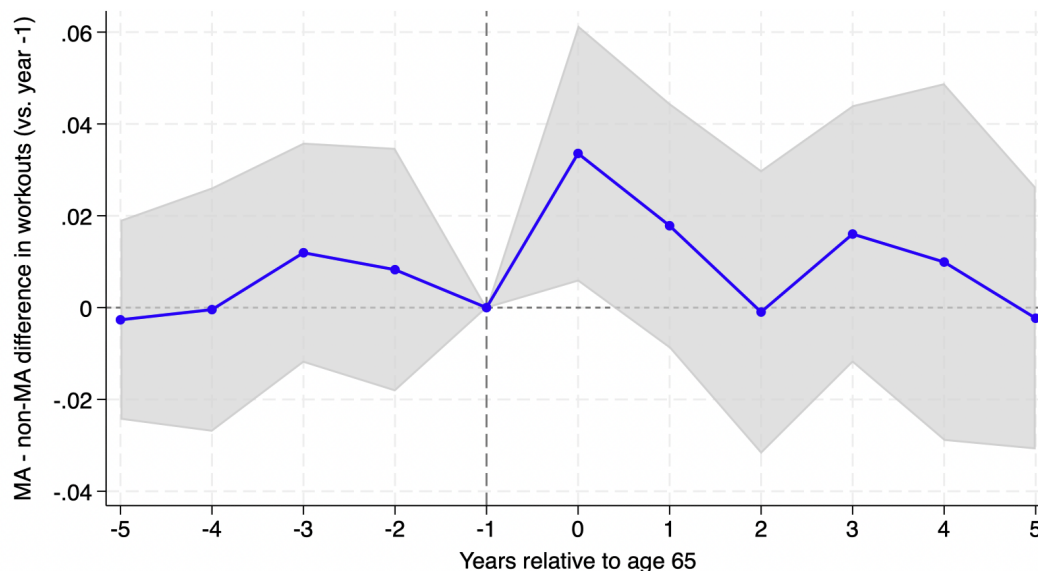


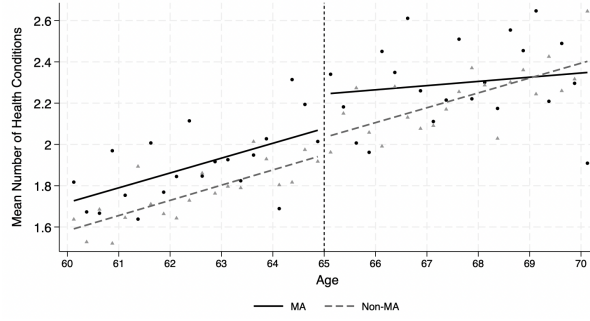
Figure A1

Figure A1: Event-Study Estimates of Daily Exercise Around Medicare Eligibility Points report estimated age-bin coefficients for the difference in daily vigorous exercise between the Medicare Advantage and non-MA groups, relative to the year immediately preceding Medicare eligibility. Vertical bars show 95 percent confidence intervals. The dotted horizontal line indicates zero, and the dashed vertical line marks the omitted reference period. The figure is used primarily to assess differential pre-trends. The pre-65 estimates are statistically not significant. As MA and TM enrollment are endogenous, coefficients at and after age 65 are descriptive and are not interpreted as causal dynamic treatment effects.

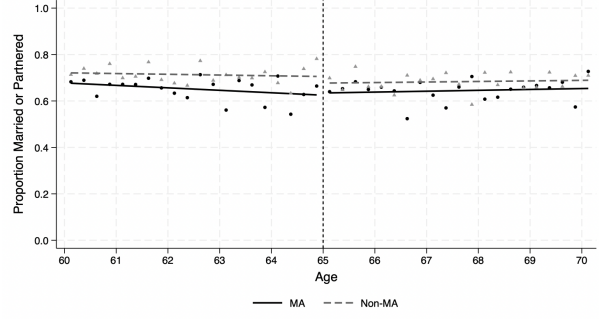
Table 4A: DiDC Estimates Across Alternative Exercise Outcomes

	Dependent Variable			
	> Once/week (1) $b = 5.000$	At least once/week (2) $b = 3.799$	Any Exercise (3) $b = 4.707$	Alternative Exercise (4) $b = 3.820$
$\Delta^{MA} - \Delta^{\text{Non-MA}}$	0.005 (0.027)	-0.004 (0.036)	-0.030 (0.030)	0.178* (0.098)
Pre-65 mean, MA group	0.276	0.394	0.514	0.658
Post-65 mean, MA group	0.281	0.386	0.491	0.706
Pre-65 mean, Non-MA group	0.306	0.415	0.536	0.711
Post-65 mean, Non-MA group	0.279	0.393	0.505	0.705
Observations	11,876	9,405	11,297	9,405
R^2	0.622	0.667	0.658	0.557
Individual fixed effects	✓	✓	✓	✓
Year fixed effects	✓	✓	✓	✓
Covariates	✓	✓	✓	✓

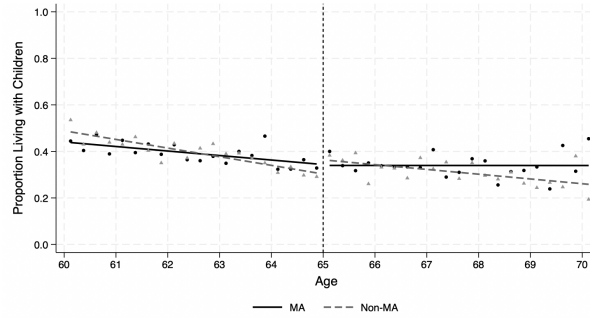
Notes: This table reports reduced-form difference-in-discontinuities (DiDC) estimates across alternative definitions of vigorous exercise. The coefficient $\Delta^{MA} - \Delta^{\text{Non-MA}}$ captures the differential discontinuity at age 65 for the MA group relative to the non-MA comparison group. The pre- and post-65 means are computed within the estimation sample and bandwidth used for each outcome. Column (1) uses an indicator equal to one if the respondent reports vigorous exercise more than once per week. Column (2) uses an indicator equal to one if the respondent reports vigorous exercise at least once per week. Column (3) uses an indicator equal to one if the respondent reports any vigorous exercise rather than never exercising. Column (4) uses an alternative workout-frequency score coded as 7 for vigorous exercise every day, 1 for vigorous exercise more than once per week, once per week, or one to three times per month, and 0 for no vigorous exercise. All specifications control for health conditions, household income, living with children, and marital status, and include individual and year fixed effects. Standard errors clustered at the individual level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.



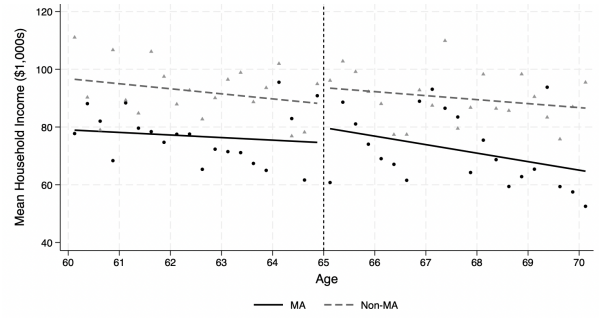
(A) Health conditions



(B) Married / partnered



(C) Living with children



(D) Household income

Figure A2

Figure A2: Points show three-month binned means for the Medicare Advantage (MA) and non-MA groups. Lines show linear fits estimated separately on each side of age 65 for each group. The vertical dashed line indicates the age-65 threshold. Panel A reports the mean number of doctor-diagnosed health conditions ranging from 0 to 8. Panel B reports the proportion of respondents who are married or partnered. Panel C reports the proportion of respondents living with at least one child. Panel D reports mean household income in thousands of dollars.